

1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4, suppress=True)

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Dispositivo: {device}')
```

Dispositivo: cuda

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints
NAMING_META = {
    'variante': 'topk',          # Arquitectura del SAE
    'tecnica': 'batch-topk',    # BatchTopK
    'capa': 6,                  # Layer del modelo OthelloGPT
    'juegos': 50000,            # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',
    'experiment_base_path': os.path.abspath('../../../experiments'),
    'extras': {
        'expansion': 8,        # d_sae / d_in = 8192 / 512
        'k': 50,               # número de features activas L0 promedio    k
        'epochs': 1000,
        'patience': 20
    }
}

print('\nMetadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')
```

Metadata de naming:

```
variante: topk
tecnica: batch-topk
capa: 6
juegos: 50000
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
```

```
extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
```

```

) if NAMING_META.get('extras') else None

# get_plots_dir crea el directorio automáticamente
plots_dir = Path(get_plots_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
))

print(f'extras_id: {extras_id}')
print(f'plots_dir: {plots_dir}')

```

Nuevo extras registrado: ex1 → {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
extras_id: ex1
plots_dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_topk_topk\50000games\ex1\plots

3. Configuración del SAE con SAELens

Configuración usando LanguageModelSAERunnerConfig de SAELens con `architecture='batchtopk'`.

Arquitectura BatchTopK: - **BatchTopK activation:** calcula un threshold global por batch para que el **promedio** de features activas por muestra sea **k** (L0 promedio **k**, pero varía por muestra) - **Sin coeficiente de sparsidad:** la sparsity se controla directamente con **k** - **Loss = MSE únicamente** (sin penalización L1)

Parámetro	L1 (standard)	TopK	BatchTopK
Sparsity	sparsity_coef (soft)	k (hard, por muestra)	k (promedio del batch)
L0	Variable	Fijo = k por muestra	Variable por muestra, promedio k
Loss	MSE + $\cdot$ L1	MSE	MSE

Nota sobre el runner: `LanguageModelSAETrainingRunner(cfg).run()` requiere que Othello-GPT esté cargado vía `TransformerLens`. Las celdas siguientes entrenan el SAE directamente sobre los archivos `.npy` pre-extraídos usando la arquitectura `BatchTopK` definida en el config.

```

NUM_EPOCHS = NAMING_META['extras']['epochs']

cfg = {
    'architecture':      'batchtopk',
    'd_in':              512,
    'd_sae':             4096,
    'k':                 50,
    'lr':                1e-3,
    'train_batch_size_tokens': 4096,
    'hook_name':         'blocks.5.hook_resid_post',
}

cfg_summary = pd.DataFrame({
    'Parámetro SAELens': ['architecture', 'd_in', 'd_sae', 'k (L0 promedio)', 'hook_name', 'lr'],
    'Valor': [cfg['architecture'], cfg['d_in'], cfg['d_sae'], cfg['k'],
              cfg['hook_name'], cfg['lr']]
})

print('Configuración SAELens:')
print(cfg_summary.to_string(index=False))

```

Configuración SAELens:

Parámetro SAELens	Valor
architecture	batchtopk
d_in	512
d_sae	4096
k (L0 promedio)	50
hook_name	blocks.5.hook_resid_post
lr	0.001

4. Dataset de Activaciones (Train/Validation Split 80-20%)

Activaciones pre-extraídas con `sae/activations/run_extraction.py` desde la capa `blocks.5.hook_resid_post`.

```

class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""
    def __init__(self, activations_path):
        self.activations = np.load(activations_path)
    def __len__(self):
        return len(self.activations)
    def __getitem__(self, idx):

```

```

        return torch.tensor(self.activations[idx], dtype=torch.float32)

activations_file = Path(f"../../activations/data/layer{NAMING_META['capa']}-{NAMING_META['id']}")
full_dataset = ActivationsDataset(activations_file)

# Split 80-20 reproducible
train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset, [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

train_loader = DataLoader(train_dataset, batch_size=cfg["train_batch_size_tokens"])
val_loader = DataLoader(val_dataset, batch_size=cfg["train_batch_size_tokens"])

dataset_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validación', 'Total'],
    'Muestras': [f'{len(train_dataset):,}', f'{len(val_dataset):,}', f'{len(full_dataset):,}'],
    'Shape': [str(full_dataset.activations.shape), '', ''],
})
print(dataset_info.to_string(index=False))

```

```

Conjunto  Muestras      Shape
Train 2,360,000 (2950000, 512)
Validación  590,000
Total 2,950,000

```

5. Modelo: BatchTopK SAE

Implementación de la arquitectura BatchTopK de SAE Lens.

Diferencia clave vs ReLU + L1 y vs TopK estándar: - El encoder proyecta a `d_sae=8192` dimensiones (pre-activación) - La activación **BatchTopK** calcula un threshold global del batch (k-ésimo valor promediado); cada muestra puede tener más o menos de `k` features activas - `L0` es **variable por muestra**, pero el **promedio del batch** `k` (soft constraint a nivel de batch) - No hay `sparsity_coef` ni cosine annealing - El decoder se normaliza a norma unitaria después de cada paso del optimizador

```

class BatchTopKSAE(nn.Module):
    """

```

```

Sparse Autoencoder con activación BatchTopK.
Equivalente a SAELens architecture='batchtopk'.

Encoder      : d_in -> d_sae (lineal + pre-bias centering)
Activación   : BatchTopK (mantiene el promedio de k features activas por batch)
Decoder      : d_sae -> d_in (columnas con norma unitaria)
"""
def __init__(self, d_in: int, d_sae: int, k: int, k_aux: int = 512):
    super().__init__()
    self.d_in = d_in
    self.d_sae = d_sae
    self.k = k
    self.k_aux = k_aux

    self.W_enc = nn.Parameter(torch.empty(d_in, d_sae))
    self.W_dec = nn.Parameter(torch.empty(d_sae, d_in))
    self.b_dec = nn.Parameter(torch.zeros(d_in))

    nn.init.kaiming_uniform_(self.W_enc)
    nn.init.kaiming_uniform_(self.W_dec)
    self.W_dec.data = self.W_dec.data / self.W_dec.data.norm(dim=-1, keepdim=True)
    self.register_buffer('num_batches_not_active', torch.zeros(d_sae))
    self.n_batches_to_dead = 10 # latente muerto si inactivo por 10 batches seguidos

def encode(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
    """Proyecta a espacio latente con activación BatchTopK"""
    x_cent = x - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    return self._topk_activation(pre_acts, threshold=threshold)

def _topk_activation(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
    """BatchTopK: selecciona exactamente k * batch_size activaciones usando scatter."""
    if threshold is None:
        acts_topk = torch.topk(x.flatten(), self.k * x.shape[0], dim=-1)
        return (
            torch.zeros_like(x.flatten())
            .scatter(-1, acts_topk.indices, acts_topk.values)
            .reshape(x.shape)
        )
    return torch.where(x > threshold, x, torch.zeros_like(x))

def decode(self, hidden: torch.Tensor) -> torch.Tensor:

```

```

        return hidden @ self.W_dec + self.b_dec

def preprocess_input(self, x: torch.Tensor):
    x_mean = x.mean(dim=-1, keepdim=True)
    x = x - x_mean
    x_std = x.std(dim=-1, keepdim=True)
    x = x / (x_std + 1e-5)
    return x, x_mean, x_std

def postprocess_output(self, x_reconstruct: torch.Tensor,
                       x_mean: torch.Tensor, x_std: torch.Tensor) -> torch.Tensor:
    return x_reconstruct * x_std + x_mean

def forward(self, x: torch.Tensor, threshold=None):
    x_norm, x_mean, x_std = self.preprocess_input(x)
    x_cent = x_norm - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    hidden = self._topk_activation(pre_acts, threshold=threshold)
    recon = self.postprocess_output(self.decode(hidden), x_mean, x_std)
    return recon, hidden, pre_acts

def auxiliary_loss(self, x: torch.Tensor, x_reconstruct: torch.Tensor, acts: torch.Tensor):
    """L_aux del repo oficial: reconstruye el residual con top-k_aux dead latents."""
    dead_features = self.num_batches_not_active >= self.n_batches_to_dead
    if dead_features.sum() == 0:
        return torch.tensor(0.0, device=x.device)
    residual = x.float() - x_reconstruct.float()
    acts_topk_aux = torch.topk(
        acts[:, dead_features],
        min(self.k_aux, int(dead_features.sum().item()))),
        dim=-1,
    )
    acts_aux = torch.zeros_like(acts[:, dead_features]).scatter(
        -1, acts_topk_aux.indices, acts_topk_aux.values
    )
    x_reconstruct_aux = acts_aux @ self.W_dec[dead_features]
    return (1/32) * (x_reconstruct_aux.float() - residual.float()).pow(2).mean()

@torch.no_grad()
def update_inactive_features(self, acts):
    self.num_batches_not_active += (acts.sum(0) == 0).float()
    self.num_batches_not_active[acts.sum(0) > 0] = 0

```

```

@torch.no_grad()
def estimate_threshold(self, loader, device, n_batches: int = 10) -> float:
    """Estima un threshold estable promediando sobre múltiples batches
    (independiente del batch actual - para usar en validación/inferencia)."""
    self.eval()
    thresholds = []
    for i, batch in enumerate(loader):
        if i >= n_batches:
            break
        x = batch.to(device)
        x_norm, _, _ = self.preprocess_input(x)
        x_cent = x_norm - self.b_dec
        pre_acts = F.relu(x_cent @ self.W_enc)
        mins = []
        for sample_idx in range(pre_acts.shape[0]):
            pos = pre_acts[sample_idx][pre_acts[sample_idx] > 0]
            if pos.numel() > 0:
                mins.append(pos.min().item())
        t = float(np.mean(mins)) if mins else 0.0
        thresholds.append(t)
    return float(np.mean(thresholds))

@torch.no_grad()
def make_decoder_weights_and_grad_unit_norm(self):
    """Proyecta el gradiente del decoder y renormaliza - igual que el repo oficial."""
    W_dec_normed = self.W_dec / self.W_dec.norm(dim=-1, keepdim=True)
    W_dec_grad_proj = (self.W_dec.grad * W_dec_normed).sum(-1, keepdim=True) * W_dec_normed
    self.W_dec.grad -= W_dec_grad_proj
    self.W_dec.data = W_dec_normed

# Crear modelo desde parámetros del cfg SAELens
model = BatchTopKSAE(d_in=cfg['d_in'], d_sae=cfg['d_sae'], k=cfg['k']).to(device)
num_params = sum(p.numel() for p in model.parameters())

model_info = pd.DataFrame({
    'Componente': [
        'Input', 'Encoder (W_enc)', 'Activación', 'Hidden activos', 'Decoder (W_dec)', 'Output'
    ],
    'k_aux': 'Total params'
},
    'Dimensión': [
        f'{cfg["d_in"]}',

```



```

        f'{cfg["d_in"]} -> {cfg["d_sae"]}',
        f'BatchTopK (k={cfg["k"]})',
        f'{cfg["d_sae"]} (solo {cfg["k"]} activos en promedio)',
        f'{cfg["d_sae"]} -> {cfg["d_in"]}',
        f'{cfg["d_in"]}',
        f'{model.k_aux}',
        f'{num_params:,'}
    ]
})
print(model_info.to_string(index=False))

```

Componente	Dimensión
Input	512
Encoder (W_enc)	512 -> 4096
Activación	BatchTopK (k=50)
Hidden activos	4096 (solo 50 activos en promedio)
Decoder (W_dec)	4096 -> 512
Output	512
k_aux	512
Total params	4,194,816

6. Función de Pérdida BatchTopK

Loss = MSE(reconstrucción)

- **MSE**: error de reconstrucción (igual que en L1)
- **Sin L1 penalty**: la sparsity ya está controlada por el threshold global de BatchTopK
- No hay `sparsity_coef` ni annealing — el único hiperparámetro de sparsity es `k` (promedio del batch)

```

ALPHA_AUX = 1 / 32 # peso de la pérdida auxiliar (arXiv:2412.06410)

def loss_function(x, reconstruction, acts, model):
    """
    Pérdida BatchTopK: MSE + alpha * L_aux (arXiv:2412.06410).
    - MSE      : error de reconstrucción
    - L_aux    : reconstruye el residual con las top-k_aux features para
                  evitar features muertas (alpha = 1/32)

    Args:
        x: Activaciones originales
    """

```

```

        reconstruction: Activaciones reconstruidas
        acts:           Pre-activaciones del encoder
        model:          Instancia de BatchTopKSAE
Returns:
    total_loss (scalar)
"""
mse    = F.mse_loss(reconstruction, x)
l_aux  = model.auxiliary_loss(x, reconstruction, acts)
return mse + l_aux

optimizer = torch.optim.Adam(model.parameters(), lr=cfg["lr"], betas=(0.9, 0.99))

optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning rate', 'betas', 'k (L0 promedio k)', 'k_aux', 'alpha_aux'],
    'Valor':      ['Adam', cfg["lr"], '(0.9, 0.99)', cfg["k"], model.k_aux, f'{ALPHA_AUX:.5f}']
})
print(optim_info.to_string(index=False))

```

Parámetro	Valor
Optimizador	Adam
Learning rate	0.001
betas	(0.9, 0.99)
k (L0 promedio k)	50
k_aux	512
alpha_aux	0.03125 (1/32)

7. Loop de Entrenamiento

```

PATIENCE = NAMING_META['extras']['patience']

save_dir = Path('./saved_model')
save_dir.mkdir(exist_ok=True)
best_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='best',
    extras_id=extras_id
)

```

```

)
BEST_CKPT = save_dir / best_model_name

history = {
    'train_mse_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_mse_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
}

best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'Iniciando entrenamiento: {NUM_EPOCHS} épocas máximo')
print(f'Early stopping: patience={PATIENCE}')
print(f'k = {cfg["k"]} features activas (L0 promedio k por batch)')
print(f'k_aux = {model.k_aux} | alpha_aux = {ALPHA_AUX:.5f} (1/32)')
print(f'Best checkpoint: {BEST_CKPT}')
print('=' * 70)

for epoch in range(NUM_EPOCHS):

    # TRAIN
    model.train()
    t_mse = t_l0 = t_frac = 0.0

    for batch in train_loader:
        x = batch.to(device)
        optimizer.zero_grad()
        recon, hidden, pre_acts = model(x)
        total_loss = loss_function(x, recon, pre_acts, model)
        total_loss.backward()
        model.make_decoder_weights_and_grad_unit_norm()
        optimizer.step()
        model.update_inactive_features(hidden)

    with torch.no_grad():
        t_mse += F.mse_loss(recon, x).item()
        t_l0 += (hidden > 0).float().sum(dim=1).mean().item()

```

```

        t_frac += (hidden > 0).float().mean().item()

n = len(train_loader)
history['train_mse_loss'].append(t_mse / n)
history['train_l0_sparsity'].append(t_l0 / n)
history['train_fraction_active'].append(t_frac / n)

# VALIDATION (sin threshold - igual que training, L0 k)
model.eval()
v_mse = v_l0 = v_frac = 0.0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)
        recon, hidden, _ = model(x)
        v_mse += F.mse_loss(recon, x).item()
        v_l0 += (hidden > 0).float().sum(dim=1).mean().item()
        v_frac += (hidden > 0).float().mean().item()

nv = len(val_loader)
history['val_mse_loss'].append(v_mse / nv)
history['val_l0_sparsity'].append(v_l0 / nv)
history['val_fraction_active'].append(v_frac / nv)

print(f"Época {epoch+1:3d}/{NUM_EPOCHS} | "
      f"Train MSE: {history['train_mse_loss'][-1]:.6f} | "
      f"Val MSE: {history['val_mse_loss'][-1]:.6f} | "
      f"Train L0: {history['train_l0_sparsity'][-1]:.1f} | "
      f"Val L0: {history['val_l0_sparsity'][-1]:.1f}")

# Early stopping
if history['val_mse_loss'][-1] < best_val_mse:
    best_val_mse = history['val_mse_loss'][-1]
    best_epoch = epoch + 1
    epochs_no_improve = 0

torch.save(model.state_dict(), BEST_CKPT)
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],

```

```

        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=cfg,
        history=history,
        optimizer=optimizer
    )
    print(f'  Mejor modelo guardado (época {best_epoch} | Val MSE: {best_val_mse:.6f})')
else:
    epochs_no_improve += 1
    if epochs_no_improve >= PATIENCE:
        print(f'\nEarly stopping en época {epoch+1} (mejor: época {best_epoch})')
        break

# Restaurar mejor modelo
model.load_state_dict(torch.load(BEST_CKPT, weights_only=True))
print(f'\nMejor época: {best_epoch} | Mejor Val MSE: {best_val_mse:.6f}')

# Estimar threshold de inferencia al final del entrenamiento
inference_threshold = model.estimate_threshold(train_loader, device, n_batches=10)
print(f'Threshold de inferencia estimado: {inference_threshold:.4f}')

```

Iniciando entrenamiento: 1000 épocas máximo

Early stopping: patience=20

k = 50 features activas (L0 promedio k por batch)

k_aux = 512 | alpha_aux = 0.03125 (1/32)

Best checkpoint: saved_model\sae_topk_batch-topk_l6_50000g_ex1_best.pt

=====

Época 1/1000 | Train MSE: 0.253137 | Val MSE: 0.191585 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 1 | Val MSE: 0.191585)

Época 2/1000 | Train MSE: 0.180659 | Val MSE: 0.174340 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 2 | Val MSE: 0.174340)

Época 3/1000 | Train MSE: 0.170420 | Val MSE: 0.169027 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 3 | Val MSE: 0.169027)

Época 4/1000 | Train MSE: 0.166244 | Val MSE: 0.165768 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 4 | Val MSE: 0.165768)
Época 5/1000 | Train MSE: 0.163603 | Val MSE: 0.163926 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 5 | Val MSE: 0.163926)
Época 6/1000 | Train MSE: 0.161986 | Val MSE: 0.162726 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 6 | Val MSE: 0.162726)
Época 7/1000 | Train MSE: 0.160925 | Val MSE: 0.162030 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 7 | Val MSE: 0.162030)
Época 8/1000 | Train MSE: 0.160152 | Val MSE: 0.161472 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 8 | Val MSE: 0.161472)
Época 9/1000 | Train MSE: 0.159595 | Val MSE: 0.160959 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 9 | Val MSE: 0.160959)
Época 10/1000 | Train MSE: 0.159167 | Val MSE: 0.160691 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 10 | Val MSE: 0.160691)
Época 11/1000 | Train MSE: 0.158824 | Val MSE: 0.160376 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 11 | Val MSE: 0.160376)
Época 12/1000 | Train MSE: 0.158547 | Val MSE: 0.160278 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 12 | Val MSE: 0.160278)
Época 13/1000 | Train MSE: 0.158310 | Val MSE: 0.159929 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 13 | Val MSE: 0.159929)
Época 14/1000 | Train MSE: 0.158082 | Val MSE: 0.159796 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 14 | Val MSE: 0.159796)

Época 15/1000 | Train MSE: 0.157913 | Val MSE: 0.159751 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 15 | Val MSE: 0.159751)
 Época 16/1000 | Train MSE: 0.157774 | Val MSE: 0.159643 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 16 | Val MSE: 0.159643)
 Época 17/1000 | Train MSE: 0.157640 | Val MSE: 0.159479 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 17 | Val MSE: 0.159479)
 Época 18/1000 | Train MSE: 0.157518 | Val MSE: 0.159317 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 18 | Val MSE: 0.159317)
 Época 19/1000 | Train MSE: 0.157403 | Val MSE: 0.159254 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 19 | Val MSE: 0.159254)
 Época 20/1000 | Train MSE: 0.157296 | Val MSE: 0.159177 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 20 | Val MSE: 0.159177)
 Época 21/1000 | Train MSE: 0.157201 | Val MSE: 0.159113 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 21 | Val MSE: 0.159113)
 Época 22/1000 | Train MSE: 0.157109 | Val MSE: 0.159070 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 22 | Val MSE: 0.159070)
 Época 23/1000 | Train MSE: 0.157040 | Val MSE: 0.158981 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 23 | Val MSE: 0.158981)
 Época 24/1000 | Train MSE: 0.156971 | Val MSE: 0.158917 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 24 | Val MSE: 0.158917)
 Época 25/1000 | Train MSE: 0.156908 | Val MSE: 0.158873 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 25 | Val MSE: 0.158873)
 Época 26/1000 | Train MSE: 0.156850 | Val MSE: 0.158803 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 26 | Val MSE: 0.158803)
 Época 27/1000 | Train MSE: 0.156795 | Val MSE: 0.158916 | Train L0: 50.0 | Val L0: 50.0
 Época 28/1000 | Train MSE: 0.156748 | Val MSE: 0.158773 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 28 | Val MSE: 0.158773)
 Época 29/1000 | Train MSE: 0.156694 | Val MSE: 0.158691 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 29 | Val MSE: 0.158691)
 Época 30/1000 | Train MSE: 0.156650 | Val MSE: 0.158703 | Train L0: 50.0 | Val L0: 50.0
 Época 31/1000 | Train MSE: 0.156604 | Val MSE: 0.158723 | Train L0: 50.0 | Val L0: 50.0
 Época 32/1000 | Train MSE: 0.156567 | Val MSE: 0.158581 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 32 | Val MSE: 0.158581)
 Época 33/1000 | Train MSE: 0.156529 | Val MSE: 0.158541 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 33 | Val MSE: 0.158541)
 Época 34/1000 | Train MSE: 0.156493 | Val MSE: 0.158538 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 34 | Val MSE: 0.158538)
 Época 35/1000 | Train MSE: 0.156461 | Val MSE: 0.158471 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 35 | Val MSE: 0.158471)
 Época 36/1000 | Train MSE: 0.156429 | Val MSE: 0.158471 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 36 | Val MSE: 0.158471)
 Época 37/1000 | Train MSE: 0.156389 | Val MSE: 0.158446 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 37 | Val MSE: 0.158446)
 Época 38/1000 | Train MSE: 0.156364 | Val MSE: 0.158471 | Train L0: 50.0 | Val L0: 50.0
 Época 39/1000 | Train MSE: 0.156329 | Val MSE: 0.158533 | Train L0: 50.0 | Val L0: 50.0
 Época 40/1000 | Train MSE: 0.156307 | Val MSE: 0.158371 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 40 | Val MSE: 0.158371)

Época 41/1000 | Train MSE: 0.156274 | Val MSE: 0.158347 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 41 | Val MSE: 0.158347)

Época 42/1000 | Train MSE: 0.156246 | Val MSE: 0.158316 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 42 | Val MSE: 0.158316)

Época 43/1000 | Train MSE: 0.156214 | Val MSE: 0.158321 | Train L0: 50.0 | Val L0: 50.0

Época 44/1000 | Train MSE: 0.156189 | Val MSE: 0.158314 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 44 | Val MSE: 0.158314)

Época 45/1000 | Train MSE: 0.156162 | Val MSE: 0.158254 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 45 | Val MSE: 0.158254)

Época 46/1000 | Train MSE: 0.156141 | Val MSE: 0.158207 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 46 | Val MSE: 0.158207)

Época 47/1000 | Train MSE: 0.156114 | Val MSE: 0.158212 | Train L0: 50.0 | Val L0: 50.0

Época 48/1000 | Train MSE: 0.156094 | Val MSE: 0.158294 | Train L0: 50.0 | Val L0: 50.0

Época 49/1000 | Train MSE: 0.156080 | Val MSE: 0.158164 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 49 | Val MSE: 0.158164)

Época 50/1000 | Train MSE: 0.156053 | Val MSE: 0.158234 | Train L0: 50.0 | Val L0: 50.0

Época 51/1000 | Train MSE: 0.156030 | Val MSE: 0.158148 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 51 | Val MSE: 0.158148)

Época 52/1000 | Train MSE: 0.156009 | Val MSE: 0.158139 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 52 | Val MSE: 0.158139)

Época 53/1000 | Train MSE: 0.155998 | Val MSE: 0.158138 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 53 | Val MSE: 0.158138)

Época 54/1000 | Train MSE: 0.155975 | Val MSE: 0.158100 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 54 | Val MSE: 0.158100)

Época 55/1000 | Train MSE: 0.155954 | Val MSE: 0.158078 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 55 | Val MSE: 0.158078)

Época 56/1000 | Train MSE: 0.155939 | Val MSE: 0.158072 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 56 | Val MSE: 0.158072)

Época 57/1000 | Train MSE: 0.155918 | Val MSE: 0.158065 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 57 | Val MSE: 0.158065)

Época 58/1000 | Train MSE: 0.155905 | Val MSE: 0.158098 | Train L0: 50.0 | Val L0: 50.0

Época 59/1000 | Train MSE: 0.155892 | Val MSE: 0.158007 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 59 | Val MSE: 0.158007)

Época 60/1000 | Train MSE: 0.155874 | Val MSE: 0.157992 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 60 | Val MSE: 0.157992)

Época 61/1000 | Train MSE: 0.155858 | Val MSE: 0.158129 | Train L0: 50.0 | Val L0: 50.0

Época 62/1000 | Train MSE: 0.155847 | Val MSE: 0.157978 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 62 | Val MSE: 0.157978)

Época 63/1000 | Train MSE: 0.155825 | Val MSE: 0.158069 | Train L0: 50.0 | Val L0: 50.0

Época 64/1000 | Train MSE: 0.155818 | Val MSE: 0.157976 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 64 | Val MSE: 0.157976)

Época 65/1000 | Train MSE: 0.155802 | Val MSE: 0.157923 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 65 | Val MSE: 0.157923)

Época 66/1000 | Train MSE: 0.155789 | Val MSE: 0.157967 | Train L0: 50.0 | Val L0: 50.0

Época 67/1000 | Train MSE: 0.155775 | Val MSE: 0.157937 | Train L0: 50.0 | Val L0: 50.0

Época 68/1000 | Train MSE: 0.155761 | Val MSE: 0.157911 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 68 | Val MSE: 0.157911)
Época 69/1000 | Train MSE: 0.155746 | Val MSE: 0.157916 | Train L0: 50.0 | Val L0: 50.0
Época 70/1000 | Train MSE: 0.155737 | Val MSE: 0.157892 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 70 | Val MSE: 0.157892)
Época 71/1000 | Train MSE: 0.155722 | Val MSE: 0.157889 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 71 | Val MSE: 0.157889)
Época 72/1000 | Train MSE: 0.155713 | Val MSE: 0.157900 | Train L0: 50.0 | Val L0: 50.0
Época 73/1000 | Train MSE: 0.155702 | Val MSE: 0.157864 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 73 | Val MSE: 0.157864)
Época 74/1000 | Train MSE: 0.155684 | Val MSE: 0.157855 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 74 | Val MSE: 0.157855)
Época 75/1000 | Train MSE: 0.155674 | Val MSE: 0.157890 | Train L0: 50.0 | Val L0: 50.0
Época 76/1000 | Train MSE: 0.155661 | Val MSE: 0.157929 | Train L0: 50.0 | Val L0: 50.0
Época 77/1000 | Train MSE: 0.155653 | Val MSE: 0.157816 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 77 | Val MSE: 0.157816)
Época 78/1000 | Train MSE: 0.155646 | Val MSE: 0.157872 | Train L0: 50.0 | Val L0: 50.0
Época 79/1000 | Train MSE: 0.155630 | Val MSE: 0.157822 | Train L0: 50.0 | Val L0: 50.0
Época 80/1000 | Train MSE: 0.155620 | Val MSE: 0.157811 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 80 | Val MSE: 0.157811)
Época 81/1000 | Train MSE: 0.155608 | Val MSE: 0.157778 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 81 | Val MSE: 0.157778)
Época 82/1000 | Train MSE: 0.155601 | Val MSE: 0.157808 | Train L0: 50.0 | Val L0: 50.0
Época 83/1000 | Train MSE: 0.155597 | Val MSE: 0.157823 | Train L0: 50.0 | Val L0: 50.0
Época 84/1000 | Train MSE: 0.155581 | Val MSE: 0.157791 | Train L0: 50.0 | Val L0: 50.0
Época 85/1000 | Train MSE: 0.155571 | Val MSE: 0.157733 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt
Mejor modelo guardado (época 85 | Val MSE: 0.157733)

Época 86/1000 | Train MSE: 0.155553 | Val MSE: 0.157782 | Train L0: 50.0 | Val L0: 50.0
 Época 87/1000 | Train MSE: 0.155544 | Val MSE: 0.157712 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 87 | Val MSE: 0.157712)
 Época 88/1000 | Train MSE: 0.155536 | Val MSE: 0.157710 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 88 | Val MSE: 0.157710)
 Época 89/1000 | Train MSE: 0.155533 | Val MSE: 0.157783 | Train L0: 50.0 | Val L0: 50.0
 Época 90/1000 | Train MSE: 0.155519 | Val MSE: 0.157683 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 90 | Val MSE: 0.157683)
 Época 91/1000 | Train MSE: 0.155514 | Val MSE: 0.157688 | Train L0: 50.0 | Val L0: 50.0
 Época 92/1000 | Train MSE: 0.155506 | Val MSE: 0.157761 | Train L0: 50.0 | Val L0: 50.0
 Época 93/1000 | Train MSE: 0.155500 | Val MSE: 0.157740 | Train L0: 50.0 | Val L0: 50.0
 Época 94/1000 | Train MSE: 0.155490 | Val MSE: 0.157723 | Train L0: 50.0 | Val L0: 50.0
 Época 95/1000 | Train MSE: 0.155481 | Val MSE: 0.157679 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 95 | Val MSE: 0.157679)
 Época 96/1000 | Train MSE: 0.155472 | Val MSE: 0.157676 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 96 | Val MSE: 0.157676)
 Época 97/1000 | Train MSE: 0.155469 | Val MSE: 0.157707 | Train L0: 50.0 | Val L0: 50.0
 Época 98/1000 | Train MSE: 0.155461 | Val MSE: 0.157660 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 98 | Val MSE: 0.157660)
 Época 99/1000 | Train MSE: 0.155449 | Val MSE: 0.157732 | Train L0: 50.0 | Val L0: 50.0
 Época 100/1000 | Train MSE: 0.155447 | Val MSE: 0.157691 | Train L0: 50.0 | Val L0: 50.0
 Época 101/1000 | Train MSE: 0.155443 | Val MSE: 0.157621 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 101 | Val MSE: 0.157621)
 Época 102/1000 | Train MSE: 0.155434 | Val MSE: 0.157631 | Train L0: 50.0 | Val L0: 50.0
 Época 103/1000 | Train MSE: 0.155430 | Val MSE: 0.157709 | Train L0: 50.0 | Val L0: 50.0
 Época 104/1000 | Train MSE: 0.155419 | Val MSE: 0.157621 | Train L0: 50.0 | Val L0: 50.0
 Época 105/1000 | Train MSE: 0.155419 | Val MSE: 0.157641 | Train L0: 50.0 | Val L0: 50.0
 Época 106/1000 | Train MSE: 0.155408 | Val MSE: 0.157744 | Train L0: 50.0 | Val L0: 50.0
 Época 107/1000 | Train MSE: 0.155405 | Val MSE: 0.157670 | Train L0: 50.0 | Val L0: 50.0

Época 108/1000 | Train MSE: 0.155395 | Val MSE: 0.157627 | Train L0: 50.0 | Val L0: 50.0
 Época 109/1000 | Train MSE: 0.155388 | Val MSE: 0.157653 | Train L0: 50.0 | Val L0: 50.0
 Época 110/1000 | Train MSE: 0.155391 | Val MSE: 0.157610 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 110 | Val MSE: 0.157610)
 Época 111/1000 | Train MSE: 0.155377 | Val MSE: 0.157586 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 111 | Val MSE: 0.157586)
 Época 112/1000 | Train MSE: 0.155374 | Val MSE: 0.157605 | Train L0: 50.0 | Val L0: 50.0
 Época 113/1000 | Train MSE: 0.155373 | Val MSE: 0.157613 | Train L0: 50.0 | Val L0: 50.0
 Época 114/1000 | Train MSE: 0.155368 | Val MSE: 0.157724 | Train L0: 50.0 | Val L0: 50.0
 Época 115/1000 | Train MSE: 0.155358 | Val MSE: 0.157601 | Train L0: 50.0 | Val L0: 50.0
 Época 116/1000 | Train MSE: 0.155354 | Val MSE: 0.157712 | Train L0: 50.0 | Val L0: 50.0
 Época 117/1000 | Train MSE: 0.155344 | Val MSE: 0.157572 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 117 | Val MSE: 0.157572)
 Época 118/1000 | Train MSE: 0.155347 | Val MSE: 0.157568 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 118 | Val MSE: 0.157568)
 Época 119/1000 | Train MSE: 0.155341 | Val MSE: 0.157560 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 119 | Val MSE: 0.157560)
 Época 120/1000 | Train MSE: 0.155334 | Val MSE: 0.157586 | Train L0: 50.0 | Val L0: 50.0
 Época 121/1000 | Train MSE: 0.155325 | Val MSE: 0.157719 | Train L0: 50.0 | Val L0: 50.0
 Época 122/1000 | Train MSE: 0.155331 | Val MSE: 0.157564 | Train L0: 50.0 | Val L0: 50.0
 Época 123/1000 | Train MSE: 0.155317 | Val MSE: 0.157577 | Train L0: 50.0 | Val L0: 50.0
 Época 124/1000 | Train MSE: 0.155313 | Val MSE: 0.157595 | Train L0: 50.0 | Val L0: 50.0
 Época 125/1000 | Train MSE: 0.155311 | Val MSE: 0.157595 | Train L0: 50.0 | Val L0: 50.0
 Época 126/1000 | Train MSE: 0.155306 | Val MSE: 0.157602 | Train L0: 50.0 | Val L0: 50.0
 Época 127/1000 | Train MSE: 0.155304 | Val MSE: 0.157550 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 127 | Val MSE: 0.157550)
 Época 128/1000 | Train MSE: 0.155297 | Val MSE: 0.157547 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 128 | Val MSE: 0.157547)
 Época 129/1000 | Train MSE: 0.155293 | Val MSE: 0.157543 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 129 | Val MSE: 0.157543)

Época 130/1000 | Train MSE: 0.155287 | Val MSE: 0.157554 | Train L0: 50.0 | Val L0: 50.0

Época 131/1000 | Train MSE: 0.155281 | Val MSE: 0.157535 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 131 | Val MSE: 0.157535)

Época 132/1000 | Train MSE: 0.155271 | Val MSE: 0.157534 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 132 | Val MSE: 0.157534)

Época 133/1000 | Train MSE: 0.155269 | Val MSE: 0.157535 | Train L0: 50.0 | Val L0: 50.0

Época 134/1000 | Train MSE: 0.155267 | Val MSE: 0.157537 | Train L0: 50.0 | Val L0: 50.0

Época 135/1000 | Train MSE: 0.155263 | Val MSE: 0.157563 | Train L0: 50.0 | Val L0: 50.0

Época 136/1000 | Train MSE: 0.155257 | Val MSE: 0.157557 | Train L0: 50.0 | Val L0: 50.0

Época 137/1000 | Train MSE: 0.155257 | Val MSE: 0.157506 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 137 | Val MSE: 0.157506)

Época 138/1000 | Train MSE: 0.155249 | Val MSE: 0.157565 | Train L0: 50.0 | Val L0: 50.0

Época 139/1000 | Train MSE: 0.155244 | Val MSE: 0.157516 | Train L0: 50.0 | Val L0: 50.0

Época 140/1000 | Train MSE: 0.155237 | Val MSE: 0.157852 | Train L0: 50.0 | Val L0: 50.0

Época 141/1000 | Train MSE: 0.155232 | Val MSE: 0.157687 | Train L0: 50.0 | Val L0: 50.0

Época 142/1000 | Train MSE: 0.155226 | Val MSE: 0.157485 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 142 | Val MSE: 0.157485)

Época 143/1000 | Train MSE: 0.155222 | Val MSE: 0.157525 | Train L0: 50.0 | Val L0: 50.0

Época 144/1000 | Train MSE: 0.155216 | Val MSE: 0.157487 | Train L0: 50.0 | Val L0: 50.0

Época 145/1000 | Train MSE: 0.155214 | Val MSE: 0.157469 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 145 | Val MSE: 0.157469)

Época 146/1000 | Train MSE: 0.155208 | Val MSE: 0.157457 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 146 | Val MSE: 0.157457)

Época 147/1000 | Train MSE: 0.155203 | Val MSE: 0.157461 | Train L0: 50.0 | Val L0: 50.0

Época 148/1000 | Train MSE: 0.155201 | Val MSE: 0.157485 | Train L0: 50.0 | Val L0: 50.0

Época 149/1000 | Train MSE: 0.155196 | Val MSE: 0.157438 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 149 | Val MSE: 0.157438)

Época 150/1000 | Train MSE: 0.155197 | Val MSE: 0.157489 | Train L0: 50.0 | Val L0: 50.0

Época 151/1000 | Train MSE: 0.155190 | Val MSE: 0.157440 | Train L0: 50.0 | Val L0: 50.0

Época 152/1000 | Train MSE: 0.155189 | Val MSE: 0.157567 | Train L0: 50.0 | Val L0: 50.0

Época 153/1000 | Train MSE: 0.155183 | Val MSE: 0.157523 | Train L0: 50.0 | Val L0: 50.0

Época 154/1000 | Train MSE: 0.155185 | Val MSE: 0.157428 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 154 | Val MSE: 0.157428)

Época 155/1000 | Train MSE: 0.155183 | Val MSE: 0.157437 | Train L0: 50.0 | Val L0: 50.0

Época 156/1000 | Train MSE: 0.155176 | Val MSE: 0.157424 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 156 | Val MSE: 0.157424)

Época 157/1000 | Train MSE: 0.155174 | Val MSE: 0.157447 | Train L0: 50.0 | Val L0: 50.0

Época 158/1000 | Train MSE: 0.155175 | Val MSE: 0.157506 | Train L0: 50.0 | Val L0: 50.0

Época 159/1000 | Train MSE: 0.155167 | Val MSE: 0.157428 | Train L0: 50.0 | Val L0: 50.0

Época 160/1000 | Train MSE: 0.155164 | Val MSE: 0.157424 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 160 | Val MSE: 0.157424)

Época 161/1000 | Train MSE: 0.155164 | Val MSE: 0.157417 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 161 | Val MSE: 0.157417)

Época 162/1000 | Train MSE: 0.155163 | Val MSE: 0.157420 | Train L0: 50.0 | Val L0: 50.0

Época 163/1000 | Train MSE: 0.155158 | Val MSE: 0.157484 | Train L0: 50.0 | Val L0: 50.0

Época 164/1000 | Train MSE: 0.155154 | Val MSE: 0.157413 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 164 | Val MSE: 0.157413)

Época 165/1000 | Train MSE: 0.155157 | Val MSE: 0.157458 | Train L0: 50.0 | Val L0: 50.0

Época 166/1000 | Train MSE: 0.155143 | Val MSE: 0.157398 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 166 | Val MSE: 0.157398)

Época 167/1000 | Train MSE: 0.155151 | Val MSE: 0.157425 | Train L0: 50.0 | Val L0: 50.0

Época 168/1000 | Train MSE: 0.155143 | Val MSE: 0.157407 | Train L0: 50.0 | Val L0: 50.0

Época 169/1000 | Train MSE: 0.155136 | Val MSE: 0.157394 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 169 | Val MSE: 0.157394)

Época 170/1000 | Train MSE: 0.155132 | Val MSE: 0.157553 | Train L0: 50.0 | Val L0: 50.0

Época 171/1000 | Train MSE: 0.155132 | Val MSE: 0.157410 | Train L0: 50.0 | Val L0: 50.0
 Época 172/1000 | Train MSE: 0.155125 | Val MSE: 0.157440 | Train L0: 50.0 | Val L0: 50.0
 Época 173/1000 | Train MSE: 0.155125 | Val MSE: 0.157410 | Train L0: 50.0 | Val L0: 50.0
 Época 174/1000 | Train MSE: 0.155117 | Val MSE: 0.157531 | Train L0: 50.0 | Val L0: 50.0
 Época 175/1000 | Train MSE: 0.155128 | Val MSE: 0.157372 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 175 | Val MSE: 0.157372)
 Época 176/1000 | Train MSE: 0.155112 | Val MSE: 0.157417 | Train L0: 50.0 | Val L0: 50.0
 Época 177/1000 | Train MSE: 0.155113 | Val MSE: 0.157453 | Train L0: 50.0 | Val L0: 50.0
 Época 178/1000 | Train MSE: 0.155110 | Val MSE: 0.157442 | Train L0: 50.0 | Val L0: 50.0
 Época 179/1000 | Train MSE: 0.155102 | Val MSE: 0.157388 | Train L0: 50.0 | Val L0: 50.0
 Época 180/1000 | Train MSE: 0.155100 | Val MSE: 0.157398 | Train L0: 50.0 | Val L0: 50.0
 Época 181/1000 | Train MSE: 0.155098 | Val MSE: 0.157397 | Train L0: 50.0 | Val L0: 50.0
 Época 182/1000 | Train MSE: 0.155095 | Val MSE: 0.157373 | Train L0: 50.0 | Val L0: 50.0
 Época 183/1000 | Train MSE: 0.155097 | Val MSE: 0.157428 | Train L0: 50.0 | Val L0: 50.0
 Época 184/1000 | Train MSE: 0.155091 | Val MSE: 0.157386 | Train L0: 50.0 | Val L0: 50.0
 Época 185/1000 | Train MSE: 0.155087 | Val MSE: 0.157366 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 185 | Val MSE: 0.157366)
 Época 186/1000 | Train MSE: 0.155085 | Val MSE: 0.157564 | Train L0: 50.0 | Val L0: 50.0
 Época 187/1000 | Train MSE: 0.155083 | Val MSE: 0.157367 | Train L0: 50.0 | Val L0: 50.0
 Época 188/1000 | Train MSE: 0.155083 | Val MSE: 0.157363 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 188 | Val MSE: 0.157363)
 Época 189/1000 | Train MSE: 0.155079 | Val MSE: 0.157373 | Train L0: 50.0 | Val L0: 50.0
 Época 190/1000 | Train MSE: 0.155081 | Val MSE: 0.157429 | Train L0: 50.0 | Val L0: 50.0
 Época 191/1000 | Train MSE: 0.155078 | Val MSE: 0.157403 | Train L0: 50.0 | Val L0: 50.0
 Época 192/1000 | Train MSE: 0.155074 | Val MSE: 0.157370 | Train L0: 50.0 | Val L0: 50.0
 Época 193/1000 | Train MSE: 0.155070 | Val MSE: 0.157350 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 193 | Val MSE: 0.157350)
 Época 194/1000 | Train MSE: 0.155069 | Val MSE: 0.157382 | Train L0: 50.0 | Val L0: 50.0
 Época 195/1000 | Train MSE: 0.155069 | Val MSE: 0.157508 | Train L0: 50.0 | Val L0: 50.0
 Época 196/1000 | Train MSE: 0.155063 | Val MSE: 0.157503 | Train L0: 50.0 | Val L0: 50.0
 Época 197/1000 | Train MSE: 0.155065 | Val MSE: 0.157396 | Train L0: 50.0 | Val L0: 50.0
 Época 198/1000 | Train MSE: 0.155061 | Val MSE: 0.157380 | Train L0: 50.0 | Val L0: 50.0
 Época 199/1000 | Train MSE: 0.155056 | Val MSE: 0.157363 | Train L0: 50.0 | Val L0: 50.0
 Época 200/1000 | Train MSE: 0.155058 | Val MSE: 0.157352 | Train L0: 50.0 | Val L0: 50.0
 Época 201/1000 | Train MSE: 0.155055 | Val MSE: 0.157386 | Train L0: 50.0 | Val L0: 50.0

Época 202/1000 | Train MSE: 0.155054 | Val MSE: 0.157348 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 202 | Val MSE: 0.157348)
 Época 203/1000 | Train MSE: 0.155057 | Val MSE: 0.157374 | Train L0: 50.0 | Val L0: 50.0
 Época 204/1000 | Train MSE: 0.155054 | Val MSE: 0.157352 | Train L0: 50.0 | Val L0: 50.0
 Época 205/1000 | Train MSE: 0.155055 | Val MSE: 0.157384 | Train L0: 50.0 | Val L0: 50.0
 Época 206/1000 | Train MSE: 0.155044 | Val MSE: 0.157344 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 206 | Val MSE: 0.157344)
 Época 207/1000 | Train MSE: 0.155044 | Val MSE: 0.157333 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 207 | Val MSE: 0.157333)
 Época 208/1000 | Train MSE: 0.155045 | Val MSE: 0.157386 | Train L0: 50.0 | Val L0: 50.0
 Época 209/1000 | Train MSE: 0.155040 | Val MSE: 0.157389 | Train L0: 50.0 | Val L0: 50.0
 Época 210/1000 | Train MSE: 0.155038 | Val MSE: 0.157363 | Train L0: 50.0 | Val L0: 50.0
 Época 211/1000 | Train MSE: 0.155037 | Val MSE: 0.157339 | Train L0: 50.0 | Val L0: 50.0
 Época 212/1000 | Train MSE: 0.155033 | Val MSE: 0.157362 | Train L0: 50.0 | Val L0: 50.0
 Época 213/1000 | Train MSE: 0.155033 | Val MSE: 0.157332 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 213 | Val MSE: 0.157332)
 Época 214/1000 | Train MSE: 0.155031 | Val MSE: 0.157363 | Train L0: 50.0 | Val L0: 50.0
 Época 215/1000 | Train MSE: 0.155032 | Val MSE: 0.157341 | Train L0: 50.0 | Val L0: 50.0
 Época 216/1000 | Train MSE: 0.155024 | Val MSE: 0.157325 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 216 | Val MSE: 0.157325)
 Época 217/1000 | Train MSE: 0.155024 | Val MSE: 0.157479 | Train L0: 50.0 | Val L0: 50.0
 Época 218/1000 | Train MSE: 0.155022 | Val MSE: 0.157400 | Train L0: 50.0 | Val L0: 50.0
 Época 219/1000 | Train MSE: 0.155010 | Val MSE: 0.157375 | Train L0: 50.0 | Val L0: 50.0
 Época 220/1000 | Train MSE: 0.155019 | Val MSE: 0.157396 | Train L0: 50.0 | Val L0: 50.0
 Época 221/1000 | Train MSE: 0.155014 | Val MSE: 0.157345 | Train L0: 50.0 | Val L0: 50.0
 Época 222/1000 | Train MSE: 0.155010 | Val MSE: 0.157329 | Train L0: 50.0 | Val L0: 50.0
 Época 223/1000 | Train MSE: 0.155009 | Val MSE: 0.157389 | Train L0: 50.0 | Val L0: 50.0
 Época 224/1000 | Train MSE: 0.155006 | Val MSE: 0.157358 | Train L0: 50.0 | Val L0: 50.0
 Época 225/1000 | Train MSE: 0.155003 | Val MSE: 0.157344 | Train L0: 50.0 | Val L0: 50.0
 Época 226/1000 | Train MSE: 0.155003 | Val MSE: 0.157475 | Train L0: 50.0 | Val L0: 50.0
 Época 227/1000 | Train MSE: 0.155002 | Val MSE: 0.157353 | Train L0: 50.0 | Val L0: 50.0
 Época 228/1000 | Train MSE: 0.155004 | Val MSE: 0.157320 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 228 | Val MSE: 0.157320)

Época 229/1000 | Train MSE: 0.154993 | Val MSE: 0.157321 | Train L0: 50.0 | Val L0: 50.0

Época 230/1000 | Train MSE: 0.154998 | Val MSE: 0.157313 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 230 | Val MSE: 0.157313)

Época 231/1000 | Train MSE: 0.154988 | Val MSE: 0.157467 | Train L0: 50.0 | Val L0: 50.0

Época 232/1000 | Train MSE: 0.154997 | Val MSE: 0.157313 | Train L0: 50.0 | Val L0: 50.0

Época 233/1000 | Train MSE: 0.154993 | Val MSE: 0.157311 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 233 | Val MSE: 0.157311)

Época 234/1000 | Train MSE: 0.154986 | Val MSE: 0.157284 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 234 | Val MSE: 0.157284)

Época 235/1000 | Train MSE: 0.154984 | Val MSE: 0.157321 | Train L0: 50.0 | Val L0: 50.0

Época 236/1000 | Train MSE: 0.154982 | Val MSE: 0.157318 | Train L0: 50.0 | Val L0: 50.0

Época 237/1000 | Train MSE: 0.154989 | Val MSE: 0.157331 | Train L0: 50.0 | Val L0: 50.0

Época 238/1000 | Train MSE: 0.154983 | Val MSE: 0.157309 | Train L0: 50.0 | Val L0: 50.0

Época 239/1000 | Train MSE: 0.154975 | Val MSE: 0.157280 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 239 | Val MSE: 0.157280)

Época 240/1000 | Train MSE: 0.154982 | Val MSE: 0.157304 | Train L0: 50.0 | Val L0: 50.0

Época 241/1000 | Train MSE: 0.154979 | Val MSE: 0.157286 | Train L0: 50.0 | Val L0: 50.0

Época 242/1000 | Train MSE: 0.154977 | Val MSE: 0.157463 | Train L0: 50.0 | Val L0: 50.0

Época 243/1000 | Train MSE: 0.154981 | Val MSE: 0.157305 | Train L0: 50.0 | Val L0: 50.0

Época 244/1000 | Train MSE: 0.154970 | Val MSE: 0.157286 | Train L0: 50.0 | Val L0: 50.0

Época 245/1000 | Train MSE: 0.154967 | Val MSE: 0.157475 | Train L0: 50.0 | Val L0: 50.0

Época 246/1000 | Train MSE: 0.154973 | Val MSE: 0.157315 | Train L0: 50.0 | Val L0: 50.0

Época 247/1000 | Train MSE: 0.154972 | Val MSE: 0.157310 | Train L0: 50.0 | Val L0: 50.0

Época 248/1000 | Train MSE: 0.154962 | Val MSE: 0.157425 | Train L0: 50.0 | Val L0: 50.0

Época 249/1000 | Train MSE: 0.154967 | Val MSE: 0.157336 | Train L0: 50.0 | Val L0: 50.0

Época 250/1000 | Train MSE: 0.154963 | Val MSE: 0.157277 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_16_50000g_ex1_best.pt

Mejor modelo guardado (época 250 | Val MSE: 0.157277)

Época 251/1000 | Train MSE: 0.154958 | Val MSE: 0.157294 | Train L0: 50.0 | Val L0: 50.0

Época 252/1000 | Train MSE: 0.154962 | Val MSE: 0.157289 | Train L0: 50.0 | Val L0: 50.0

Época 253/1000 | Train MSE: 0.154958 | Val MSE: 0.157279 | Train L0: 50.0 | Val L0: 50.0

Época 254/1000 | Train MSE: 0.154954 | Val MSE: 0.157404 | Train L0: 50.0 | Val L0: 50.0

Época 255/1000 | Train MSE: 0.154955 | Val MSE: 0.157342 | Train L0: 50.0 | Val L0: 50.0
 Época 256/1000 | Train MSE: 0.154959 | Val MSE: 0.157273 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 256 | Val MSE: 0.157273)
 Época 257/1000 | Train MSE: 0.154951 | Val MSE: 0.157327 | Train L0: 50.0 | Val L0: 50.0
 Época 258/1000 | Train MSE: 0.154947 | Val MSE: 0.157275 | Train L0: 50.0 | Val L0: 50.0
 Época 259/1000 | Train MSE: 0.154945 | Val MSE: 0.157305 | Train L0: 50.0 | Val L0: 50.0
 Época 260/1000 | Train MSE: 0.154948 | Val MSE: 0.157263 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 260 | Val MSE: 0.157263)
 Época 261/1000 | Train MSE: 0.154945 | Val MSE: 0.157308 | Train L0: 50.0 | Val L0: 50.0
 Época 262/1000 | Train MSE: 0.154943 | Val MSE: 0.157311 | Train L0: 50.0 | Val L0: 50.0
 Época 263/1000 | Train MSE: 0.154937 | Val MSE: 0.157284 | Train L0: 50.0 | Val L0: 50.0
 Época 264/1000 | Train MSE: 0.154943 | Val MSE: 0.157270 | Train L0: 50.0 | Val L0: 50.0
 Época 265/1000 | Train MSE: 0.154941 | Val MSE: 0.157276 | Train L0: 50.0 | Val L0: 50.0
 Época 266/1000 | Train MSE: 0.154942 | Val MSE: 0.157429 | Train L0: 50.0 | Val L0: 50.0
 Época 267/1000 | Train MSE: 0.154940 | Val MSE: 0.157318 | Train L0: 50.0 | Val L0: 50.0
 Época 268/1000 | Train MSE: 0.154936 | Val MSE: 0.157264 | Train L0: 50.0 | Val L0: 50.0
 Época 269/1000 | Train MSE: 0.154937 | Val MSE: 0.157287 | Train L0: 50.0 | Val L0: 50.0
 Época 270/1000 | Train MSE: 0.154934 | Val MSE: 0.157398 | Train L0: 50.0 | Val L0: 50.0
 Época 271/1000 | Train MSE: 0.154936 | Val MSE: 0.157248 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 271 | Val MSE: 0.157248)
 Época 272/1000 | Train MSE: 0.154931 | Val MSE: 0.157293 | Train L0: 50.0 | Val L0: 50.0
 Época 273/1000 | Train MSE: 0.154932 | Val MSE: 0.157280 | Train L0: 50.0 | Val L0: 50.0
 Época 274/1000 | Train MSE: 0.154927 | Val MSE: 0.157441 | Train L0: 50.0 | Val L0: 50.0
 Época 275/1000 | Train MSE: 0.154931 | Val MSE: 0.157256 | Train L0: 50.0 | Val L0: 50.0
 Época 276/1000 | Train MSE: 0.154924 | Val MSE: 0.157262 | Train L0: 50.0 | Val L0: 50.0
 Época 277/1000 | Train MSE: 0.154924 | Val MSE: 0.157286 | Train L0: 50.0 | Val L0: 50.0
 Época 278/1000 | Train MSE: 0.154923 | Val MSE: 0.157254 | Train L0: 50.0 | Val L0: 50.0
 Época 279/1000 | Train MSE: 0.154921 | Val MSE: 0.157277 | Train L0: 50.0 | Val L0: 50.0
 Época 280/1000 | Train MSE: 0.154922 | Val MSE: 0.157412 | Train L0: 50.0 | Val L0: 50.0
 Época 281/1000 | Train MSE: 0.154924 | Val MSE: 0.157295 | Train L0: 50.0 | Val L0: 50.0
 Época 282/1000 | Train MSE: 0.154917 | Val MSE: 0.157262 | Train L0: 50.0 | Val L0: 50.0
 Época 283/1000 | Train MSE: 0.154917 | Val MSE: 0.157283 | Train L0: 50.0 | Val L0: 50.0
 Época 284/1000 | Train MSE: 0.154916 | Val MSE: 0.157314 | Train L0: 50.0 | Val L0: 50.0
 Época 285/1000 | Train MSE: 0.154914 | Val MSE: 0.157219 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 285 | Val MSE: 0.157219)

Época 286/1000 | Train MSE: 0.154913 | Val MSE: 0.157223 | Train L0: 50.0 | Val L0: 50.0
 Época 287/1000 | Train MSE: 0.154909 | Val MSE: 0.157218 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 287 | Val MSE: 0.157218)
 Época 288/1000 | Train MSE: 0.154912 | Val MSE: 0.157248 | Train L0: 50.0 | Val L0: 50.0
 Época 289/1000 | Train MSE: 0.154908 | Val MSE: 0.157239 | Train L0: 50.0 | Val L0: 50.0
 Época 290/1000 | Train MSE: 0.154907 | Val MSE: 0.157245 | Train L0: 50.0 | Val L0: 50.0
 Época 291/1000 | Train MSE: 0.154905 | Val MSE: 0.157216 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 291 | Val MSE: 0.157216)
 Época 292/1000 | Train MSE: 0.154900 | Val MSE: 0.157242 | Train L0: 50.0 | Val L0: 50.0
 Época 293/1000 | Train MSE: 0.154901 | Val MSE: 0.157214 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 293 | Val MSE: 0.157214)
 Época 294/1000 | Train MSE: 0.154897 | Val MSE: 0.157247 | Train L0: 50.0 | Val L0: 50.0
 Época 295/1000 | Train MSE: 0.154897 | Val MSE: 0.157194 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 295 | Val MSE: 0.157194)
 Época 296/1000 | Train MSE: 0.154894 | Val MSE: 0.157197 | Train L0: 50.0 | Val L0: 50.0
 Época 297/1000 | Train MSE: 0.154893 | Val MSE: 0.157308 | Train L0: 50.0 | Val L0: 50.0
 Época 298/1000 | Train MSE: 0.154890 | Val MSE: 0.157215 | Train L0: 50.0 | Val L0: 50.0
 Época 299/1000 | Train MSE: 0.154891 | Val MSE: 0.157257 | Train L0: 50.0 | Val L0: 50.0
 Época 300/1000 | Train MSE: 0.154894 | Val MSE: 0.157248 | Train L0: 50.0 | Val L0: 50.0
 Época 301/1000 | Train MSE: 0.154893 | Val MSE: 0.157190 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 301 | Val MSE: 0.157190)
 Época 302/1000 | Train MSE: 0.154885 | Val MSE: 0.157217 | Train L0: 50.0 | Val L0: 50.0
 Época 303/1000 | Train MSE: 0.154889 | Val MSE: 0.157243 | Train L0: 50.0 | Val L0: 50.0
 Época 304/1000 | Train MSE: 0.154888 | Val MSE: 0.157343 | Train L0: 50.0 | Val L0: 50.0
 Época 305/1000 | Train MSE: 0.154886 | Val MSE: 0.157208 | Train L0: 50.0 | Val L0: 50.0
 Época 306/1000 | Train MSE: 0.154883 | Val MSE: 0.157276 | Train L0: 50.0 | Val L0: 50.0
 Época 307/1000 | Train MSE: 0.154882 | Val MSE: 0.157222 | Train L0: 50.0 | Val L0: 50.0
 Época 308/1000 | Train MSE: 0.154887 | Val MSE: 0.157330 | Train L0: 50.0 | Val L0: 50.0
 Época 309/1000 | Train MSE: 0.154880 | Val MSE: 0.157350 | Train L0: 50.0 | Val L0: 50.0
 Época 310/1000 | Train MSE: 0.154882 | Val MSE: 0.157204 | Train L0: 50.0 | Val L0: 50.0
 Época 311/1000 | Train MSE: 0.154878 | Val MSE: 0.157223 | Train L0: 50.0 | Val L0: 50.0
 Época 312/1000 | Train MSE: 0.154875 | Val MSE: 0.157242 | Train L0: 50.0 | Val L0: 50.0
 Época 313/1000 | Train MSE: 0.154878 | Val MSE: 0.157236 | Train L0: 50.0 | Val L0: 50.0

Época 314/1000 | Train MSE: 0.154877 | Val MSE: 0.157206 | Train L0: 50.0 | Val L0: 50.0
 Época 315/1000 | Train MSE: 0.154875 | Val MSE: 0.157214 | Train L0: 50.0 | Val L0: 50.0
 Época 316/1000 | Train MSE: 0.154871 | Val MSE: 0.157255 | Train L0: 50.0 | Val L0: 50.0
 Época 317/1000 | Train MSE: 0.154871 | Val MSE: 0.157199 | Train L0: 50.0 | Val L0: 50.0
 Época 318/1000 | Train MSE: 0.154869 | Val MSE: 0.157182 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 318 | Val MSE: 0.157182)
 Época 319/1000 | Train MSE: 0.154871 | Val MSE: 0.157195 | Train L0: 50.0 | Val L0: 50.0
 Época 320/1000 | Train MSE: 0.154870 | Val MSE: 0.157223 | Train L0: 50.0 | Val L0: 50.0
 Época 321/1000 | Train MSE: 0.154868 | Val MSE: 0.157239 | Train L0: 50.0 | Val L0: 50.0
 Época 322/1000 | Train MSE: 0.154866 | Val MSE: 0.157267 | Train L0: 50.0 | Val L0: 50.0
 Época 323/1000 | Train MSE: 0.154867 | Val MSE: 0.157216 | Train L0: 50.0 | Val L0: 50.0
 Época 324/1000 | Train MSE: 0.154866 | Val MSE: 0.157187 | Train L0: 50.0 | Val L0: 50.0
 Época 325/1000 | Train MSE: 0.154863 | Val MSE: 0.157330 | Train L0: 50.0 | Val L0: 50.0
 Época 326/1000 | Train MSE: 0.154858 | Val MSE: 0.157255 | Train L0: 50.0 | Val L0: 50.0
 Época 327/1000 | Train MSE: 0.154865 | Val MSE: 0.157387 | Train L0: 50.0 | Val L0: 50.0
 Época 328/1000 | Train MSE: 0.154866 | Val MSE: 0.157197 | Train L0: 50.0 | Val L0: 50.0
 Época 329/1000 | Train MSE: 0.154860 | Val MSE: 0.157209 | Train L0: 50.0 | Val L0: 50.0
 Época 330/1000 | Train MSE: 0.154858 | Val MSE: 0.157335 | Train L0: 50.0 | Val L0: 50.0
 Época 331/1000 | Train MSE: 0.154854 | Val MSE: 0.157269 | Train L0: 50.0 | Val L0: 50.0
 Época 332/1000 | Train MSE: 0.154861 | Val MSE: 0.157238 | Train L0: 50.0 | Val L0: 50.0
 Época 333/1000 | Train MSE: 0.154855 | Val MSE: 0.157244 | Train L0: 50.0 | Val L0: 50.0
 Época 334/1000 | Train MSE: 0.154854 | Val MSE: 0.157170 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 334 | Val MSE: 0.157170)
 Época 335/1000 | Train MSE: 0.154861 | Val MSE: 0.157244 | Train L0: 50.0 | Val L0: 50.0
 Época 336/1000 | Train MSE: 0.154850 | Val MSE: 0.157367 | Train L0: 50.0 | Val L0: 50.0
 Época 337/1000 | Train MSE: 0.154854 | Val MSE: 0.157197 | Train L0: 50.0 | Val L0: 50.0
 Época 338/1000 | Train MSE: 0.154852 | Val MSE: 0.157188 | Train L0: 50.0 | Val L0: 50.0
 Época 339/1000 | Train MSE: 0.154848 | Val MSE: 0.157169 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt
 Mejor modelo guardado (época 339 | Val MSE: 0.157169)
 Época 340/1000 | Train MSE: 0.154848 | Val MSE: 0.157197 | Train L0: 50.0 | Val L0: 50.0
 Época 341/1000 | Train MSE: 0.154850 | Val MSE: 0.157251 | Train L0: 50.0 | Val L0: 50.0
 Época 342/1000 | Train MSE: 0.154851 | Val MSE: 0.157200 | Train L0: 50.0 | Val L0: 50.0
 Época 343/1000 | Train MSE: 0.154849 | Val MSE: 0.157214 | Train L0: 50.0 | Val L0: 50.0
 Época 344/1000 | Train MSE: 0.154843 | Val MSE: 0.157200 | Train L0: 50.0 | Val L0: 50.0
 Época 345/1000 | Train MSE: 0.154843 | Val MSE: 0.157236 | Train L0: 50.0 | Val L0: 50.0
 Época 346/1000 | Train MSE: 0.154843 | Val MSE: 0.157220 | Train L0: 50.0 | Val L0: 50.0
 Época 347/1000 | Train MSE: 0.154838 | Val MSE: 0.157370 | Train L0: 50.0 | Val L0: 50.0

Época 348/1000	Train MSE: 0.154845	Val MSE: 0.157325	Train L0: 50.0	Val L0: 50.0
Época 349/1000	Train MSE: 0.154837	Val MSE: 0.157216	Train L0: 50.0	Val L0: 50.0
Época 350/1000	Train MSE: 0.154835	Val MSE: 0.157186	Train L0: 50.0	Val L0: 50.0
Época 351/1000	Train MSE: 0.154833	Val MSE: 0.157330	Train L0: 50.0	Val L0: 50.0
Época 352/1000	Train MSE: 0.154839	Val MSE: 0.157171	Train L0: 50.0	Val L0: 50.0
Época 353/1000	Train MSE: 0.154839	Val MSE: 0.157301	Train L0: 50.0	Val L0: 50.0
Época 354/1000	Train MSE: 0.154836	Val MSE: 0.157188	Train L0: 50.0	Val L0: 50.0
Época 355/1000	Train MSE: 0.154838	Val MSE: 0.157181	Train L0: 50.0	Val L0: 50.0
Época 356/1000	Train MSE: 0.154833	Val MSE: 0.157198	Train L0: 50.0	Val L0: 50.0
Época 357/1000	Train MSE: 0.154830	Val MSE: 0.157277	Train L0: 50.0	Val L0: 50.0
Época 358/1000	Train MSE: 0.154831	Val MSE: 0.157160	Train L0: 50.0	Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 358 | Val MSE: 0.157160)

Época 359/1000	Train MSE: 0.154834	Val MSE: 0.157183	Train L0: 50.0	Val L0: 50.0
Época 360/1000	Train MSE: 0.154826	Val MSE: 0.157340	Train L0: 50.0	Val L0: 50.0
Época 361/1000	Train MSE: 0.154828	Val MSE: 0.157223	Train L0: 50.0	Val L0: 50.0
Época 362/1000	Train MSE: 0.154829	Val MSE: 0.157178	Train L0: 50.0	Val L0: 50.0
Época 363/1000	Train MSE: 0.154827	Val MSE: 0.157199	Train L0: 50.0	Val L0: 50.0
Época 364/1000	Train MSE: 0.154829	Val MSE: 0.157196	Train L0: 50.0	Val L0: 50.0
Época 365/1000	Train MSE: 0.154825	Val MSE: 0.157163	Train L0: 50.0	Val L0: 50.0
Época 366/1000	Train MSE: 0.154823	Val MSE: 0.157317	Train L0: 50.0	Val L0: 50.0
Época 367/1000	Train MSE: 0.154826	Val MSE: 0.157167	Train L0: 50.0	Val L0: 50.0
Época 368/1000	Train MSE: 0.154821	Val MSE: 0.157192	Train L0: 50.0	Val L0: 50.0
Época 369/1000	Train MSE: 0.154828	Val MSE: 0.157189	Train L0: 50.0	Val L0: 50.0
Época 370/1000	Train MSE: 0.154822	Val MSE: 0.157192	Train L0: 50.0	Val L0: 50.0
Época 371/1000	Train MSE: 0.154823	Val MSE: 0.157198	Train L0: 50.0	Val L0: 50.0
Época 372/1000	Train MSE: 0.154821	Val MSE: 0.157151	Train L0: 50.0	Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\50000games\sae_topk_batch-topk_l6_50000g_ex1_best.pt

Mejor modelo guardado (época 372 | Val MSE: 0.157151)

Época 373/1000	Train MSE: 0.154820	Val MSE: 0.157205	Train L0: 50.0	Val L0: 50.0
Época 374/1000	Train MSE: 0.154818	Val MSE: 0.157178	Train L0: 50.0	Val L0: 50.0
Época 375/1000	Train MSE: 0.154821	Val MSE: 0.157173	Train L0: 50.0	Val L0: 50.0
Época 376/1000	Train MSE: 0.154820	Val MSE: 0.157179	Train L0: 50.0	Val L0: 50.0
Época 377/1000	Train MSE: 0.154816	Val MSE: 0.157267	Train L0: 50.0	Val L0: 50.0
Época 378/1000	Train MSE: 0.154819	Val MSE: 0.157174	Train L0: 50.0	Val L0: 50.0
Época 379/1000	Train MSE: 0.154812	Val MSE: 0.157188	Train L0: 50.0	Val L0: 50.0
Época 380/1000	Train MSE: 0.154813	Val MSE: 0.157204	Train L0: 50.0	Val L0: 50.0
Época 381/1000	Train MSE: 0.154815	Val MSE: 0.157316	Train L0: 50.0	Val L0: 50.0
Época 382/1000	Train MSE: 0.154814	Val MSE: 0.157292	Train L0: 50.0	Val L0: 50.0
Época 383/1000	Train MSE: 0.154812	Val MSE: 0.157366	Train L0: 50.0	Val L0: 50.0
Época 384/1000	Train MSE: 0.154812	Val MSE: 0.157318	Train L0: 50.0	Val L0: 50.0

Época 385/1000	Train MSE: 0.154811	Val MSE: 0.157311	Train L0: 50.0	Val L0: 50.0
Época 386/1000	Train MSE: 0.154815	Val MSE: 0.157201	Train L0: 50.0	Val L0: 50.0
Época 387/1000	Train MSE: 0.154812	Val MSE: 0.157219	Train L0: 50.0	Val L0: 50.0
Época 388/1000	Train MSE: 0.154812	Val MSE: 0.157193	Train L0: 50.0	Val L0: 50.0
Época 389/1000	Train MSE: 0.154811	Val MSE: 0.157179	Train L0: 50.0	Val L0: 50.0
Época 390/1000	Train MSE: 0.154809	Val MSE: 0.157209	Train L0: 50.0	Val L0: 50.0
Época 391/1000	Train MSE: 0.154808	Val MSE: 0.157189	Train L0: 50.0	Val L0: 50.0
Época 392/1000	Train MSE: 0.154810	Val MSE: 0.157165	Train L0: 50.0	Val L0: 50.0

Early stopping en época 392 (mejor: época 372)

Mejor época: 372 | Mejor Val MSE: 0.157151

Threshold de inferencia estimado: 0.0015

8. Guardar Modelo

```
# Guardar checkpoint final
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': {
        'architecture': cfg["architecture"],
        'd_in': cfg["d_in"],
        'd_sae': cfg["d_sae"],
        'k': cfg["k"],
        'lr': cfg["lr"],
        'hook_name': cfg["hook_name"],
    },
    'history': history,
}

final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras_id=extras_id
)
checkpoint_path = save_dir / final_model_name
torch.save(checkpoint, checkpoint_path)
```

```

print(f'Modelo guardado: {checkpoint_path}')

save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras=NAMING_META['extras'],
    config=cfg,
    history=history,
    optimizer=optimizer
)

```

Modelo guardado: saved_model\sae_topk_batch-topk_l6_50000g_ex1_final.pt

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_batch-topk_l6_50000games\sae_topk_batch-topk_l6_50000g_ex1_final.pt

WindowsPath('C:/Users/Esposa/Documents/Repos/sae-othello-gpt/layer_06/models/sae_topk_batch-topk_l6_50000games/sae_topk_batch-topk_l6_50000g_ex1_final.pt')

9. Métricas básicas del modelo

```

import json

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

training_metrics = {
    # Calidad de reconstrucción
    'val_mse_final': history['val_mse_loss'][-1],
    'val_mse_best': best_val_mse,
}

```



```

'val_l0_final':          history['val_l0_sparsity'][-1],
'val_fraction_active_final': history['val_fraction_active'][-1],
# Contexto del entrenamiento
'best_epoch':           best_epoch,
'total_epochs':         len(history['train_mse_loss']),
'train_mse_final':      history['train_mse_loss'][-1],
'train_l0_final':       history['train_l0_sparsity'][-1],
# Hiperparámetros TopK
'architecture':         cfg["architecture"],
'k':                    cfg["k"],
'd_sae':                cfg["d_sae"],
'd_in':                 cfg["d_in"],
'expansion_factor':     cfg["d_sae"] // cfg["d_in"],
'hook_name':            cfg["hook_name"],
}

os.makedirs(metrics_dir, exist_ok=True)
metrics_path = Path(metrics_dir) / 'training_metrics.json'
with open(metrics_path, 'w') as f:
    json.dump(training_metrics, f, indent=2)

print(f'Métricas guardadas: {metrics_path}')
print(pd.DataFrame(
    list(training_metrics.items()), columns=['Métrica', 'Valor']
).to_string(index=False))

```

Métricas guardadas: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae\topk_topk\50000games\ex1\metrics\training_metrics.json

Métrica	Valor
val_mse_final	0.157165
val_mse_best	0.157151
val_l0_final	50.0
val_fraction_active_final	0.012207
best_epoch	372
total_epochs	392
train_mse_final	0.15481
train_l0_final	50.0
architecture	batchtopk
k	50
d_sae	4096
d_in	512
expansion_factor	8

```
hook_name blocks.5.hook_resid_post
```

10. Visualización de Resultados

```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# MSE Loss
axes[0].plot(history['train_mse_loss'], label='Train', color='steelblue', alpha=0.8)
axes[0].plot(history['val_mse_loss'], label='Validación', color='tomato', alpha=0.8)
if best_epoch > 0:
    axes[0].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7,
                    label=f'Best (época {best_epoch})')
axes[0].set_title('MSE Loss (Reconstrucción)')
axes[0].set_xlabel('Época')
axes[0].set_ylabel('MSE')
axes[0].legend()
axes[0].grid(True, alpha=0.3)

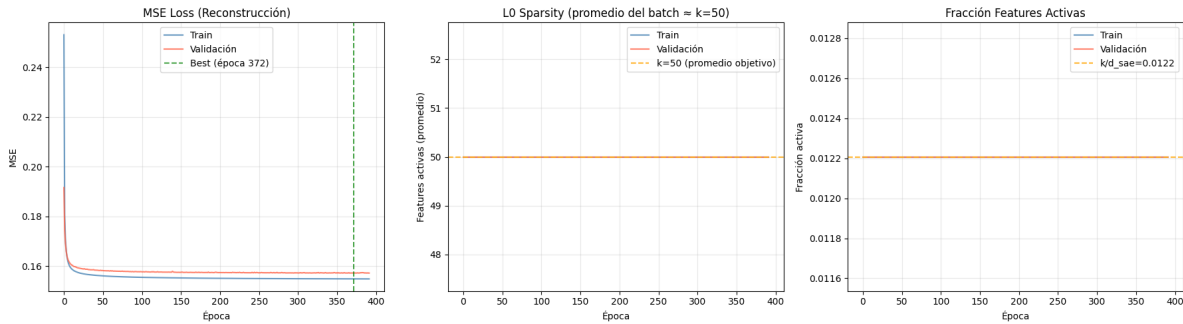
# L0 sparsity (para BatchTopK el promedio del batch k, pero varía por muestra)
axes[1].plot(history['train_l0_sparsity'], label='Train', color='steelblue', alpha=0.8)
axes[1].plot(history['val_l0_sparsity'], label='Validación', color='tomato', alpha=0.8)
axes[1].axhline(cfg["k"], color='orange', linestyle='--', alpha=0.8, label=f'k={cfg["k"]} (p)')
axes[1].set_title(f'L0 Sparsity (promedio del batch k={cfg["k"]})')
axes[1].set_xlabel('Época')
axes[1].set_ylabel('Features activas (promedio)')
axes[1].legend()
axes[1].grid(True, alpha=0.3)

# Fracción activa
axes[2].plot(history['train_fraction_active'], label='Train', color='steelblue', alpha=0.8)
axes[2].plot(history['val_fraction_active'], label='Validación', color='tomato', alpha=0.8)
expected_frac = cfg["k"] / cfg["d_sae"]
axes[2].axhline(expected_frac, color='orange', linestyle='--', alpha=0.8,
                label=f'k/d_sae={expected_frac:.4f}')
axes[2].set_title('Fracción Features Activas')
axes[2].set_xlabel('Época')
axes[2].set_ylabel('Fracción activa')
axes[2].legend()
axes[2].grid(True, alpha=0.3)

plt.tight_layout()
```

```
plt.savefig(plots_dir / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('Gráficas guardadas exitosamente')
print('='*50)
```



```
=====
Gráficas guardadas exitosamente
=====
```

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```
import subprocess

exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
```

```

    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_batch-topk.ipynb'
output_path = exp_dir / pdf_name

print(f'Generando PDF con Quarto...')
print(f'Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f'PDF generado exitosamente: {output_path}')
else:
    print(f'Error al generar PDF:')
    print(result.stderr)

```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk_topk\50000games\ex1\sae_topk_batch-topk_l6_50000g_ex1_training.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk_topk\50000games\ex1\sae_topk_batch-topk_l6_50000g_ex1_training.pdf